VW1321 Energy Tee Long

Paper Packaging - Contact Discoloration Investigation

Technical Review and Possible Root Cause Action - Hypothesis

NG Description and Observation Summary

A localized discoloration mark was observed on the front of the white garment, primarily in areas reported to have direct contact with the paper made Vela Bag.

Product Background

Style	VW1321 Energy Tee in WHT color
Article / Composition	KEH125 / 96% Polyester + 4% Elastane
T1/T2 (Vertical)	JJAMM Works International
Manufacturer	Nantong Kfine Intl Co., Ltd., China

Phenomenon and Observation

A noticeable discoloration mark was observed on the front side of the garment.

The affected pattern corresponds to locations reported to have direct contact with the paper made Vela Bag, while non-contact areas did not show the same visible staining mark.



Figure 1. Initial garment condition and Vela Bag contact reference

Experiment by Home Laundry Response: Before VS. After



Figure 2. Visual comparison Before and After one standard home-laundry cycle

Key evidence: After one standard home-laundry cycle, the discoloration mark became much less noticeable, while the overall whiteness appeared less bright. This observation suggests that laundering may have altered or removed surface-associated materials. The wash response is consistent with a reversible surface discoloration mechanism.

Interpretation: The wash response is more consistent with a removable surface residue, contamination, or reversible chemical interaction than with an irreversible color change. The specific chemical pathway such as Optical Brightener (OB/OBA), finishing chemicals, or other removable surface residues require further analytical confirmation.

Observation under the UV fluorescence - Affected contact area VS. Non-contact area

The Affected contact area seems wearing a mask – the staining mark is blocking the brightness.

Key evidence: A localized reduction in UV fluorescence was observed in the affected contact area. This may be associated with OBA masking, alteration, degradation or partial loss.

Additional analytical testing is required to distinguish among these mechanisms.

3 x SGS Test Reports Overview – from Manufacturer side

Test Item	Report # SL326033538456O1TX (Sep 15 2026)	Report # SL326033539056O1TX (Sep 16 2026)	Report # SL326033539355O1TX (Sep 28 2026)
Colorfastness to Dye Transfer to Storage	Pass	Pass	Pass
Colorfastness to Perspiration	Pass	Pass	N/A
Colorfastness to Phenolic Yellowing	N/A	Fail (Grade: 4) VS. Requirement: 4.5	Fail (Grade: 4) VS. Requirement: 4.5



Figure 3. Summary of the three SGS submissions and reported colorfastness results

Interpretation: These three SGS submissions/reports should be considered as supporting test history, the failed results of Phenolic Yellowing at Grade at 4.0 would be considered as Marginally Acceptable, in addition to the observed garment mark described in this study as discoloration because the visible staining became less non-noticeable after one standard home laundering.

Additional Verification: 5 Packaging and Garment Test Reports

Five additional laboratory reports were reviewed to verify BHT content and to evaluate Phenolic Yellowing and Heat-air ageing under the specified test conditions. The results are summarized below.

No.	Report / Lab	Specimen (s)	Test / Condition	Result	RCA relevance
1	A2260858489101001 / CTI	Vela bag: white paper with black ink	BHT; US EPA 3550C / 8270E; GC-MS	N.D. (<0.5 mg/kg)	BHT not detected in this submitted packaging sample.
2	SL92629354129101TX / SGS	Paper bag A, whole bag with black print and adhesive	BHT; ASTM D4275-17; GC-MS	N.D.; Pass (RL 5 mg/kg; requirement 25 mg/kg)	Independent BHT result also does not support a BHT-driven pathway for the tested bag.
3	SL92629354128701TX / SGS	Paper bag A + white T-shirt B	Phenolic Yellowing; Courtaulds Appendix A	A: 4-5; B: 4-5; Pass (minimum 4.0)	No phenolic-yellowing failure reproduced under this test condition.
4	SL92629354129301TX / SGS	Paper bag + white T-shirt purchased from Shanghai store	Heat-air ageing; FZ/T 01008-2008; 50°C, Method B, 48 h	Colour change 4-5; no appearance/other change; Pass	Bag/garment combination met the specified heat-ageing requirement.
5	SL92629354129601TX / SGS	White T-shirt	Heat-air ageing; FZ/T 01008-2008; 50°C, Method B, 48 h	Colour change 4-5; no appearance/other change; Pass	Garment control met the same heat-ageing requirement.

Summary of 5 additional laboratory reports reviewed for the RCA update

Interpretation: The two BHT analyses reported N.D. at different reporting/detection limits:

Two packaging BHT reports: CTI Report A2260858489101001 tested white paper with black ink from the Vela bag by GC-MS and reported BHT = N.D. with an MDL of 0.5 mg/kg.

SGS Report SL92629354129101TX tested the whole paper bag with black print and adhesive by GC-MS and reported BHT = N.D. with a reporting limit of 5 mg/kg; conclusion: PASS against the stated requirement of 25 mg/kg.

The Phenolic Yellowing test Passed for both the paper bag and white T-shirt at Grade 4-5, and both Heat-air ageing reports passed with Grade 4-5 colour change and no reported appearance or other changes.

That said, these five reports do not reproduce a BHT-related, Phenolic Yellowing, or general Heat-ageing failure in the submitted samples under the specified laboratory conditions.

BHT Content Testing at SGS TPE Lab. – Verification of Seaman/Vela Bag from China

BHT Content Testing Method: ASTM D4275 (Solvent extraction followed by gas chromatographic analysis). Sampling from inside of Vela Bag where exactly had prolonged contact with the garment. Acceptable Criteria: N.D.

Textile Laboratory Amendment

Test Report No: TXA1076/2026/LI Date: Oct. 19, 2026 Page 4 of 6

Component List / List of Materials

Sample No.	Material No.	Component Description	Material	Color	Fiber Type*	Remark
/	1	White Vela Bag (Inside)	Paper	White	-	-

* This fiber type identification is for the selection of azo dye testing procedure.

Note:
Type A = Textile contains natural fibers only and/or regenerated fibers exclude the group of acetate.
Type B = Textile contains polyester fibers only
Type C = Textile contains man-made fibers and/or man-made blended fibers (except polyester fibers)

Test Result :

Butylated Hydroxytoluene (BHT)

Test Method: With reference to ASTM D4275:2017. Analysis was conducted by GC-MS.

		Result
	CAS-No.	1
Butylated Hydroxytoluene (BHT)	128-37-0	ND
Conclusion	--	**DATA**

Note: ND = Not Detected
mg/kg = ppm
Reporting Limit = 1 mg/kg

Interpretation: BHT was not detected in submitted Vela Bag sample and based on the available evidence, a BHT-driven or BHT + NOx discoloration pathway from the tested Vela Bag is not supported as the primary cause of the observed mark/discoloration.

RCA limitation: These results apply only to the tested samples and conditions, do not identify the material responsible for the original localized contact mark, and not distinguish OBA masking, OBA alteration, finishing residue, or another removable surface interaction. The original wash response and localized UV-fluorescence difference, therefore remain important evidence for the next verification stage.

Possible Root Cause Hypotheses

- **Packaging-contact interaction:** Direct and prolonged contact between the folded garment and the paper-based Vela bag may have contributed to the localized mark.
- **BHT / NOx-related discoloration – Not Supported:** BHT was reported as N.D. in the tested inside area of the Vela Bag under SGS Report TXA1076/2026/LI. Based on this result, BHT from the tested packaging sample and a BHT + NOx-related discoloration mechanism are not supported as the cause for this case.
- **Fabric-surface chemical interaction:** Possible Optical Brightener (OB/OBA) masking, alteration, degradation or partial loss, pH imbalance from the dyeing and finishing process, finishing-chemical residue, or other removable surface materials may have interacted with the Vela Bag contact area and contributed to the localized discoloration.

Recommended Verification Plan

1. BHT verification completed: CTI Report A2260858489101001 and SGS Report SL92629354129101TX both reported BHT = N.D. in the tested packaging samples. No further BHT-focused investigation is prioritized unless a different packaging source, untested component, or new contradictory evidence is identified.
2. Compare affected contact zones with non-contact zones using targeted chemical-residue screening.
3. Review fabric finishing records, including Optical Brightener use, final pH, washing / neutralization, and surface residues.
4. Conduct a controlled packing trial using the same garment with: Vela Bag, alternative paper bag, inert barrier, and unpackaged control.
5. Assess samples before laundering and again after a controlled home-laundry cycle to confirm reversibility and recurrence.
6. Compare affected and non-affected areas before and after laundering using CIE Whiteness Index and UV fluorescence intensity. Where practical, conduct targeted OBA/finishing-residue analysis to distinguish masking, alteration, degradation or partial loss.

Current Conclusion

The evidence supports a wash-reversible, localized packaging-contact discoloration mechanism. The five additional reports did not reproduce BHT contamination, phenolic-yellowing failure, or heat-air ageing failure in the submitted samples under the specified test conditions.

BHT was N.D. in both tested packaging samples, so BHT from the tested packaging and a BHT + NOx pathway should be removed from the primary RCA hypotheses. The phenolic-yellowing and heat-air ageing results were also compliant. The remaining focus is a localized packaging-contact interaction with fabric-surface chemistry, particularly OBA behavior and removable finishing or surface residues.

Possible Detection

Two Detection Methods at OQC Stage of T2/Fabric Supplier or at IQC Stage of T1/Garment Factory

Method 1: Accelerated Heat Ageing Test

Procedure:
Using heat-air ageing conditions as a reference, place randomly selected white fabric samples in a laboratory oven under controlled conditions, for example, **50°C for 48 hours**.

Purpose:
To screen the incoming fabric for potential discoloration risks associated with **optical brightening agents (OBA)** or **residual chemicals from scouring and finishing processes**. The test is intended to determine whether heat exposure may trigger **localized yellowing/discoloration, OBA degradation, or changes in OBA performance** before the fabric enters garment production.

Method 2: Simulated Packaging Contact and Pressure Test

Procedure:
Cut the IQC-selected white fabric into test specimens and place them in direct contact with the **inner contact surface of the actual Vela Bag paper packaging used for bulk production**. Apply controlled heat and pressure, for example, **50°C for 48 hours with a 1 kg load**, to simulate elevated temperature and stacking pressure that may occur during storage and container transportation.

Purpose:
To assess, before cutting and garment production, whether the incoming fabric lot is susceptible to **surface discoloration or OBA masking effects** when in direct contact with the paper packaging under accelerated heat and pressure conditions.